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Dose-Response and Normal Tissue Complication Probabilities after Proton Therapy for Choroidal Melanoma

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Purpose: Normal tissue complication probability (NTCP) models could aid the understanding of dose dependence of radiation-induced toxicities after eye-preserving radiotherapy of choroidal melanomas. We performed NTCP-modeling and established dose-response relationships for visual acuity (VA) deterioration and common late complications after treatments with proton therapy (PT).

Design: Retrospective study from single, large referral center.

Participants: We considered patients from Nice, France, diagnosed with choroidal melanoma and treated primarily with hypofractionated PT (52 Gy physical dose in 4 fractions). Complete VA deterioration information was available for 1020 patients, and complete information on late complications was available for 991 patients.

Methods: Treatment details, dose-volume histograms (DVHs) for relevant anatomic structures, and patient and tumor characteristics were available from a dedicated ocular database. Least absolute shrinkage and selection operator (LASSO) variable selection was used to identify variables with the strongest impact on each end point, followed by multivariate Cox regressions and logistic regressions to analyze the relationships among dose, clinical characteristics, and clinical outcomes.

Main Outcome Measures: Dose-response relationship for VA deterioration and late complications.

Results: Dose metrics for several structures (i.e., optic disc, macula, retina, globe, lens, ciliary body) correlated with clinical outcome. The near-maximum dose to the macula showed the strongest correlation with VA deterioration. The near-maximum dose to the retina was the only variable with clear impact on the risk of maculopathy, the dose to 20% of the optic disc had the largest impact on optic neuropathy, dose to 20% of cornea had the largest impact on neovascular glaucoma, and dose to 20% of the ciliary body had the largest impact on ocular hypertension. The volume of the ciliary body receiving 26 Gy was the only variable associated with the risk of cataract, and the volume of retina receiving 52 Gy was associated with the risk of retinal detachment. Optic disc-to-tumor distance was the only variable associated with dry eye syndrome in the absence of DVH for the lachrymal gland.

Conclusions: VA deterioration and specific late complications demonstrated dependence on dose delivered to normal structures in the eye after PT for choroidal melanoma. VA deterioration depended on dose to a range of structures, whereas more specific complications were related to dose metrics for specific structures. *Ophthalmology 2021;128:152-161 © 2020 by the American Academy of Ophthalmology*

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Eye-preserving proton therapy (PT) is used commonly to treat choroidal melanomas.[1] The ultimate objective of the treatment is to destroy the malignancy without producing complications on adjacent healthy tissues, thereby preserving long-term function. Nonetheless, some structures may be exposed to large doses during treatment, and radiation-induced visual acuity (VA) deterioration and toxicities are common side effects.[2] Normal tissue complication probability (NTCP) models for most complications have not yet been established for hypofractionated PT, and the relevance of damage to the various ocular structures has not been examined fully. To rectify this, we examined data from a large cohort of patients treated at a dedicated PT facility with long follow-up. We conducted dose-response modelling to examine the relationships between dose delivered to healthy tissue and the occurrence of VA deterioration and various radiation-induced toxicities.

Methods

Patients

This retrospective study included patients from Nice, France treated for choroidal melanomas between 1991 and 2015 at a

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